6.10.2.1 N-Channel Depletion MOSFET

Figure 6.13 shows the construction and symbol of N-channel depletion MOSFET. It consists of a lightly doped p-type semiconductor substrate. In the substrate, two highly doped N+ regions are diffused. One region is source (*S*) and other is drain (*D*). A lightly doped N-channel is formed between source and drain by diffusion. A thin layer of SiO_2 is grown over its surface. Then, an aluminium layer is formed over SiO_2, which acts as gate (*G*). SiO_2 acts as insulator. These setup acts as a parallel plate capacitor.

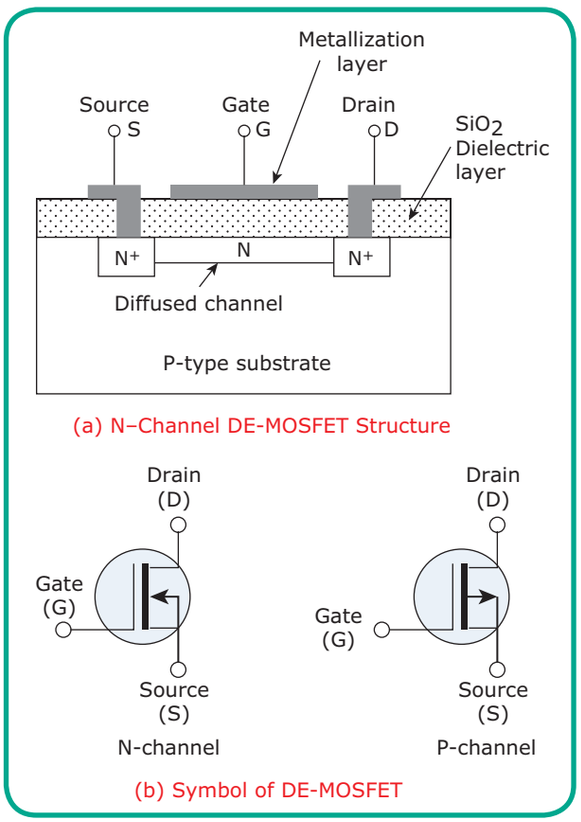


Figure 6.13 Construction and symbol of N-Channel and p-Channel Depletion MOSFET

Working Principle: The depletion MOSFET can be operated in two different modes as given below

Depletion Mode: Figure 6.14 shows the n-channel depletion mode operation. In this mode, the gate is maintained at negative potential and drain is at positive potential with respect to source. The negative voltage on the gate attracts the holes in the P-type substrate. Thus, the holes are moved towards the gate terminal. Because of the SiO_2 (insulator) layer, the holes are induced in the N-channel, which reduces the charge carriers (electrons) in the N-channel. So, the conductivity decreases, the value of drain current also decreases.

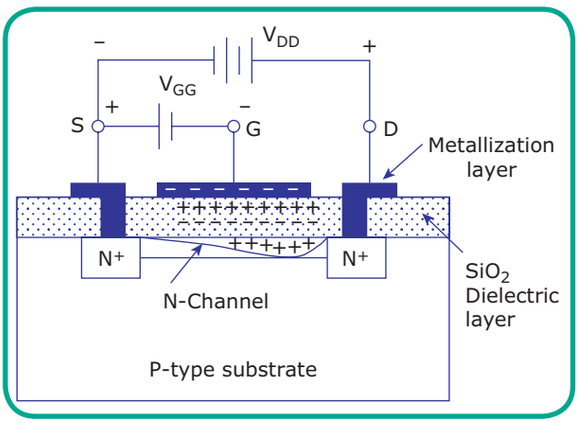


Figure 6.14 Operation of N-Channel DE-MOSFET

Enhancement Mode: In this mode, both the gate and drain are maintained at positive potential with respect to source. The positive gate voltage induces the charges in the N-channel. These induced negative charges make N-channel between source and drain more conductive. As the positive gate to source voltage increases, the drain current also increases. Hence, the mode of operation is called 'enhancement mode'.

Application

1. MOSFET is widely used for switching and amplifying the signals.
2. It is used in FM radio and TV receivers (for mixer operation).
3. It is used in computer memories.
4. It is used as auto intensity control of street lights.

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